

Bluetooth Channel Sounding

Mode-0 Frequency Estimation

A self-contained derivation from oscillator coordinates and received I/Q samples to $f_{RX}[k]$, FFO, and PBR phase calibration

This report defines every mathematical symbol and reference frame before use. It derives

the received Mode-0 complex samples from the passband RF signal, derives two frequency estimators, reconstructs the Bluetooth quantity $f_{RX}[k]$, and substitutes that result into the exact FAE-adjusted fractional frequency offset equation. A numerical example demonstrates every intermediate value and sign. The separate Volume 1 reciprocal PCT phase-calibration and Inline PCT Transfer procedures are then connected to the Mode-0 frequency result.

Primary references	Core 6.3, Vol. 1, Part A, Secs. 9.1-9.2; Vol. 6, Part A, Secs. 3.5 and 6.3
Air-interface timing	Core 6.3, Vol. 6, Part H, Secs. 4.3.1 and 4.6
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Abstract

Bluetooth Channel Sounding Mode-0 provides the relative frequency and timing calibration for the remaining steps of a CS subevent. The reflector transmits an 80 μ s unmodulated tone; the initiator estimates its average frequency, denoted $f_{RX}[k]$, and combines it with the reflector's communicated Mode-0 frequency actuation error (FAE) to estimate fractional frequency offset (FFO). The specification defines the signal interval and the FFO quantities but does not prescribe a digital frequency estimator.

The central difficulty is reference-frame bookkeeping. A receiver does not observe RF frequency directly. It observes complex phase rotation relative to its own zero-baseband frequency and measures time with its own sample clock. This report establishes a receiver-local coordinate, defines local frequency actuation error (LFAE) as an error in that zero-baseband reference, derives the complex baseband model, and proves the reconstruction

$$\widehat{f}_{RX}[k] = f_{c,cmd}^{(I)}[k] + \delta f_L^{(I)}[k] + s_{IQ} \widehat{\Delta f}_{BB}[k].$$

Substitution into the Bluetooth FFO equation gives a direct expression from I/Q phase rate to $\widehat{\text{FFO}}_{RX}[k]$. The report also derives the exact relative-oscillator interpretation, estimator ambiguity and variance, the complete 2,440 MHz example, and the distinct reciprocal PCT phase-calibration procedure from Volume 1, Part A, Sec. 9.2.

Normative boundary. Bluetooth defines $f_{RX}[k]$ as the receiving device's estimate of the average received Mode-0 tone frequency, requires compensation of local actuation error, and defines the FAE-adjusted FFO equation. The RF architecture, I/Q convention, windowing, estimator, and quality gates are implementation choices and must be validated end to end.

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1 Scope and questions answered

Mode-0 is not a distance measurement. It establishes the relative frequency and timing relationship that later RTT and PBR steps require. This report answers the following engineering questions.

1. Which part of the Mode-0 waveform is measured?
2. What exactly do $f_0[k]$, $f[k, 1]$, $f_{RX}[k]$, FAE, LFAE, and FFO mean?
3. In whose frequency coordinate is $f_{RX}[k]$ expressed?
4. How does an RF sinusoid become $I[n] + jQ[n]$?
5. How does phase rotation in radians per sample become an RF-frequency estimate in hertz?
6. How is that estimate inserted into the Bluetooth FFO equation without double-counting FAE or LFAE?
7. How are several Mode-0 results combined, and how does the result interact with PBR phase calibration?

The derivation assumes the initiator is the Mode-0 receiver and the reflector is the Mode-0 transmitter, matching the usual procedure. The equations remain valid with roles exchanged if all role labels and local calibrations are exchanged consistently.

2 Normative map and calibration layers

Core 6.3 Volume 1, Part A, Sec. 9.1 states that Mode-0 calibrates one device to the other in frequency and timing. Volume 6, Part H, Sec. 4.3.1 defines the Mode-0 packet and tone timing. Volume 6, Part A, Sec. 3.5.1 defines FFO and $f_{RX}[k]$. Volume 1, Part A, Sec. 9.2 separately defines reciprocal phase calibration for PBR.

Table 1: Normative map.

Source	Principal content	Use in this report
Vol. 1, Part A, Sec. 9.1	CS architecture and purpose of Modes 0-3	Separates frequency/time calibration from RTT and PBR measurements.
Vol. 1, Part A, Sec. 9.2	Reciprocal PCT equations and IPT architecture	Defines cancellation of relative LO phase.
Vol. 6, Part H, Sec. 4.3.1	Mode-0 sequence, T_{RD} , T_{IP1} , T_{GD} , T_{FM}	Defines the waveform interval used for frequency estimation.
Vol. 6, Part A, Sec. 3.5.1	$f_0[k]$, $f[k, p]$, FAE, $f_{RX}[k]$, $FFO_{RX}[k]$, FFO_{RX}^*	Provides the exact specification-level frequency quantities.
Vol. 6, Part A, Sec. 3.5.2	Expected later transmitted frequencies	Connects Mode-0 with later CS steps.
Vol. 6, Part A, Secs. 6.3-6.5	LFAE phase correction, PCT timing, IPT	Connects local actuation error and phase processing.
Vol. 6, Part H, Sec. 4.6	PCT measurement period and exclusion regions	Defines samples valid for phase measurement.
RFPHY.TS, Sec. 4.8.1	Instrument measurement of average Mode-0 tone frequency	Confirms the FAE-adjusted FFO calculation used for RF testing.

2.1 Three calibration layers

Layer	Observable	Function
Mode-0 frequency/time	fre- $f_{RX}[k]$, $FFO_{RX}[k]$, FFO_{RX}^*	Establishes the relative frequency scale; supports frequency placement and timing-scale compensation.
PBR phase/LO	PCT_{REFL} and PCT_{INIT} , or IPT equivalent	Cancels relative LO phase and preserves twice the reciprocal channel phase.
Local RF implementation	LFAE, sample-rate, mixer sign, I/Q and RF-path corrections	Removes local deterministic errors not represented by the peer-to-peer equations.

3 Mode-0 timing and measurement interval

3.1 Air-interface sequence

The reflector's frequency-measurement tone follows its synchronization packet and guard interval. The fixed estimator-related durations are $T_{RD} = 5 \mu\text{s}$, $T_{GD} = 10 \mu\text{s}$, and $T_{FM} = 80 \mu\text{s}$. The configured T_{IP1} lies in the allowed set 10, 20, 30, 40, 50, 60, 80, and 145 μs .

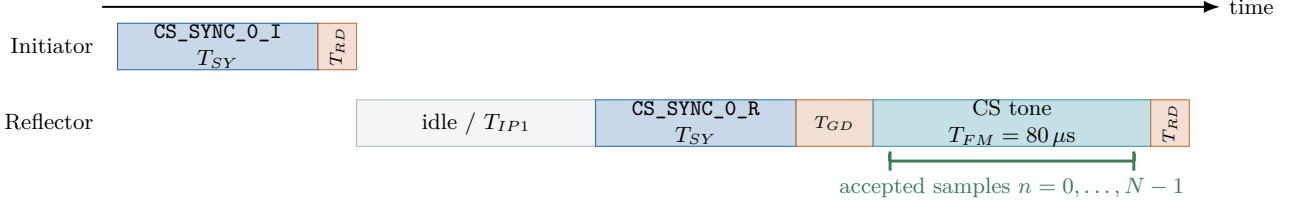


Figure 1: Mode-0 timing. Frequency is estimated from the reflector's unmodulated CS tone during T_{FM} . Widths are labeled, not proportional.

Here T_{SY} is the duration of the selected CS synchronization packet. The complete step duration is

$$T_{step,0} = 2T_{SY} + 2T_{RD} + T_{IP1} + T_{GD} + T_{FM}. \quad (1)$$

3.2 Implementation window

Let $t_{FM,0}$ be the predicted start of the received tone in the initiator's local time coordinate. A receiver can select

$$W_k = [t_{FM,0} + t_{settle}, t_{FM,0} + T_{FM} - t_{tail}], \quad (2)$$

where $t_{settle} \geq 0$ rejects front-end or NCO transients and $t_{tail} \geq 0$ rejects ramp-down contamination. Define the accepted duration and sample count as

$$T_{obs} = T_{FM} - t_{settle} - t_{tail}, \quad N = \lfloor T_{obs} F_s \rfloor. \quad (3)$$

Here $\lfloor x \rfloor$ is the greatest integer not exceeding x . The margins are implementation choices. Every reported estimate should retain N and the actual window bounds.

4 End-to-end processing overview

Figure 2 gives the complete data dependency before the detailed derivation. No block can be omitted without changing the meaning of the result.

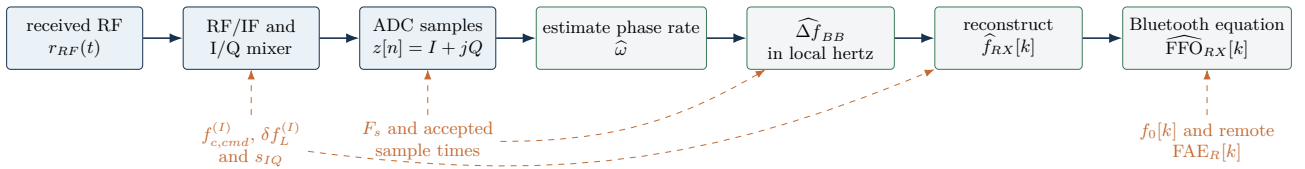


Figure 2: End-to-end Mode-0 measurement chain. Superscript (I) means a quantity expressed in the initiator's calibrated local coordinate.

5 Mathematical framework and complete symbol definitions

This section fixes the reference frame, dimensions, and sign convention used in Sections 6-8. Symbols are not used before they are defined.

5.1 Indices, devices, and time coordinates

- M is the number of Mode-0 steps at the start of the subevent, and $k \in \{1, \dots, M\}$ is a Mode-0 step index.
- R denotes the reflector, which transmits the measured Mode-0 tone.
- I denotes the initiator, which receives the tone and computes FFO.
- t is an ideal physical time coordinate used only to derive the relative-oscillator interpretation.
- τ_I is the initiator-local calibrated time coordinate used by the receiver implementation.
- $n \in \{0, \dots, N-1\}$ is a complex-sample index and $\tau_n = nT_s$.
- F_s is the calibrated complex sample rate in samples per initiator-local second; $T_s = 1/F_s$.

Let the initiator clock scale be $1 + \epsilon_I$ relative to ideal time:

$$\tau_I = (1 + \epsilon_I)t. \quad (4)$$

If a physical sinusoid is $e^{j2\pi f^{(phys)}t}$, then in initiator time

$$e^{j2\pi f^{(phys)}t} = e^{j2\pi f^{(I)}\tau_I}, \quad f^{(I)} \equiv \frac{f^{(phys)}}{1 + \epsilon_I}. \quad (5)$$

Thus a superscript (I) means “numerical frequency measured per initiator-local second.” This is the natural coordinate of its LO, sample clock, timestamps, and digital estimator. For samples generated by the same local timebase, $\tau_n = n/F_s$ and the corresponding ideal physical time is $t_n = n/[F_s(1 + \epsilon_I)]$.

5.2 Protocol frequencies and dimensionless errors

Table 2: Frequency and error symbols.

Symbol	Definition	Unit
$f_0[k]$	Nominal center frequency assigned to the CS channel in step k .	Hz
$f_R^{tx,phys}[k]$	Actual average physical RF frequency transmitted by the reflector during the Mode-0 tone.	Hz
$f_{RX}^{(I)}[k]$	Average received tone frequency expressed in the initiator-local coordinate after local calibration; this is the implementation realization of the specification's $f_{RX}[k]$.	Hz
$a_R[k]$	Dimensionless reflector FAE, $a_R[k] = 10^{-6}\text{FAE}_R[k]$.	1
$\eta_{R/I}[k]$	Dimensionless remote-to-local fractional frequency offset, $\eta_{R/I} = 10^{-6}\text{FFO}_{R/I}$.	1
$f_{ref,R}[k]$	FAE-adjusted reflector reference, $f_0[k](1 + a_R[k])$.	Hz
$f_{c,cmd}^{(I)}[k]$	Receiver command frequency that would map to zero complex-baseband frequency if local actuation were perfect. It includes any declared analog LO and digital NCO translation.	Hz
$\delta f_L^{(I)}[k]$	Local zero-baseband frequency error: actual calibrated zero-baseband reference minus commanded zero-baseband reference. Positive means the actual zero is above the command.	Hz
$f_c^{(I)}[k]$	Actual calibrated zero-baseband reference, $f_{c,cmd}^{(I)}[k] + \delta f_L^{(I)}[k]$.	Hz
$\lambda_I[k]$	Dimensionless local actuation error under this sign convention; $\delta f_L^{(I)} = f_{c,cmd}^{(I)}\lambda_I$. If stored in ppm, $\lambda_I = 10^{-6}\text{LFAE}_I$.	1
$\Delta f_{BB}[k]$	Signed rotation frequency of the accepted complex baseband tone.	Hz
s_{IQ}	I/Q orientation sign, +1 or -1, defined by Eq. 6.	1

5.3 Complex-sample and estimator symbols

Table 3: Signal-processing symbols.

Symbol	Definition	Unit
$z[n] = I[n] + jQ[n]$	Accepted complex baseband sample.	ADC units
$I[n], Q[n]$	Real in-phase and quadrature ADC components.	ADC units
j	Imaginary unit, $j^2 = -1$.	1
$\rho[n]$	Nonnegative desired-tone magnitude at sample n .	ADC units
ϕ_0	Constant received phase at the start of the accepted window.	rad
$\psi[n]$	Residual time-varying phase from phase noise, motion, or channel evolution.	rad
$d[n]$	Complex DC/LO-leakage component.	ADC units
$v[n]$	Additive complex noise and interference.	ADC units
ω	Desired mean discrete-time phase increment, $2\pi\Delta f_{BB}/F_s$.	rad/sample
m	Positive integer correlation lag.	samples
w_n	Nonnegative estimator weight.	1
R_m	Weighted lag- m complex correlation.	ADC units squared
$\text{Arg}(\cdot)$	Principal argument in $(-\pi, \pi]$.	rad
$\mathbb{E}[\cdot]$	Statistical expectation operator.	varies
$\text{var}(\cdot)$	Statistical variance operator.	squared units
q	Integer alias index used to unwrap $m\omega$.	1
$\hat{x}$	Estimate of quantity x .	same as x

5.4 I/Q sign convention

The report defines s_{IQ} through

$$\Delta f_{BB}[k] = s_{IQ} \left(f_{RX}^{(I)}[k] - f_c^{(I)}[k] \right), \quad s_{IQ} \in \{+1, -1\}. \quad (6)$$

For the common convention in which an RF tone above the zero-baseband reference rotates counterclockwise, $s_{IQ} = +1$. Swapping I and Q, conjugating samples, or using the opposite complex mixer changes the sign. Because $s_{IQ}^{-1} = s_{IQ}$, inversion gives

$$f_{RX}^{(I)}[k] = f_c^{(I)}[k] + s_{IQ}\Delta f_{BB}[k]. \quad (7)$$

5.5 Assumptions used for derivations

The exact Bluetooth FFO definitions do not require these assumptions. The I/Q estimator derivations assume: (i) the accepted tone is dominated by one narrowband component; (ii) F_s is known in the initiator coordinate; (iii) the desired phase is approximately affine over T_{obs} ; (iv) weights are real and nonnegative; and (v) the acquisition stage resolves correlation aliases. Departures from these assumptions are treated as error sources, not hidden inside undefined terms.

6 Exact FFO equations and oscillator-frame derivation

6.1 Bluetooth FAE-adjusted definition

Define the reflector FAE-adjusted reference

$$f_{ref,R}[k] = f_0[k] (1 + a_R[k]) = f_0[k] (1 + 10^{-6} \text{FAE}_R[k]). \quad (8)$$

For the actual average transmitted tone frequency, the specification-level FFO is

$$\text{FFO}_R[k] = 10^6 \frac{f_R^{tx,phys}[k] - f_{ref,R}[k]}{f_{ref,R}[k]} \text{ ppm}. \quad (9)$$

The RF conformance test uses the same form with the instrument's measured average tone frequency.

The receiving-device form is

$$\widehat{\text{FFO}}_{RX}[k] = 10^6 \frac{\widehat{f}_{RX}^{(I)}[k] - f_{ref,R}[k]}{f_{ref,R}[k]} \text{ ppm}, \quad (10)$$

where $\widehat{f}_{RX}^{(I)}[k]$ is the local-calibrated estimate derived below. Equation 10 is the endpoint. FAE is not subtracted again after this equation.

6.2 Exact relative-oscillator interpretation

Let the reflector clock scale be $1 + \epsilon_R$ and the initiator clock scale be $1 + \epsilon_I$ relative to ideal time. Factor the reflector's physical average tone frequency as

$$f_R^{tx,phys}[k] = f_0[k](1 + a_R[k])(1 + \epsilon_R). \quad (11)$$

This factorization is exact if ϵ_R is defined by Eq. 11; it does not assume that circuit mechanisms are statistically independent.

By Eq. 5, the initiator expresses that received tone as

$$f_{RX}^{(I)}[k] = \frac{f_R^{tx,phys}[k]}{1 + \epsilon_I} = f_0[k](1 + a_R[k]) \frac{1 + \epsilon_R}{1 + \epsilon_I}. \quad (12)$$

Insert Eq. 12 into Eq. 10 with a perfect estimate:

$$10^{-6} \text{FFO}_{RX}[k] = \frac{f_0(1 + a_R)(1 + \epsilon_R)/(1 + \epsilon_I) - f_0(1 + a_R)}{f_0(1 + a_R)} \quad (13)$$

$$= \frac{1 + \epsilon_R}{1 + \epsilon_I} - 1 = \frac{\epsilon_R - \epsilon_I}{1 + \epsilon_I}. \quad (14)$$

Thus Mode-0 measures the reflector's frequency scale relative to the initiator's scale after the reflector FAE and the initiator local calibration are handled. For small errors,

$$10^{-6} \text{FFO}_{RX}[k] \approx \epsilon_R - \epsilon_I. \quad (15)$$

Equivalently, define

$$\eta_{R/I}[k] \equiv \frac{1 + \epsilon_R}{1 + \epsilon_I} - 1. \quad (16)$$

Then $f_{RX}^{(I)}[k] = f_{ref,R}[k](1 + \eta_{R/I}[k])$ and $\text{FFO}_{RX}[k] = 10^6 \eta_{R/I}[k]$ ppm. This is the direct bridge between the oscillator model and the Bluetooth normalization.

6.3 Reconstruction of $f_{RX}[k]$ from baseband

The actual local zero-baseband reference is

$$f_c^{(I)}[k] = f_{c,cmd}^{(I)}[k] + \delta f_L^{(I)}[k]. \quad (17)$$

Combining Eqs. 7 and 17 gives the exact receiver architecture equation

$$f_{RX}^{(I)}[k] = f_{c,cmd}^{(I)}[k] + \delta f_L^{(I)}[k] + s_{IQ} \Delta f_{BB}[k]. \quad (18)$$

After estimation,

$$\widehat{f}_{RX}^{(I)}[k] = f_{c,cmd}^{(I)}[k] + \widehat{\delta f}_L^{(I)}[k] + s_{IQ}\widehat{\Delta f}_{BB}[k]. \quad (19)$$

This equation explains precisely where LFAE enters: it corrects the frequency represented by zero complex-baseband rotation. If a vendor table defines LFAE as command minus actual rather than actual minus command, its stored value must be negated once before use in Eq. 19.

6.4 Direct equation from measured phase rate to FFO

For a lag- m phase estimator, Section 8 derives

$$\widehat{\Delta f}_{BB}[k] = \frac{F_s}{2\pi m} (\text{Arg } R_m + 2\pi q), \quad (20)$$

where $q \in \mathbb{Z}$ is selected by coarse acquisition. Substitute Eq. 20 into Eq. 19, then into Eq. 10:

$$\widehat{\text{FFO}}_{RX}[k] = 10^6 \left[\frac{f_{c,cmd}^{(I)}[k] + \widehat{\delta f}_L^{(I)}[k] + s_{IQ} \frac{F_s}{2\pi m} (\text{Arg } R_m + 2\pi q)}{f_0[k](1 + 10^{-6}\text{FAE}_R[k])} - 1 \right] \text{ ppm}. \quad (21)$$

Every term in Eq. 21 has been defined and has units of hertz inside the ratio, except the dimensionless signs, phase, and error factors.

6.5 Useful special case

If the receiver commands zero baseband at the nominal channel, $f_{c,cmd}^{(I)}[k] = f_0[k]$, then

$$\widehat{\text{FFO}}_{RX}[k] = 10^6 \left[\frac{f_0[k] + \widehat{\delta f}_L^{(I)}[k] + s_{IQ}\widehat{\Delta f}_{BB}[k]}{f_0[k](1 + a_R[k])} - 1 \right]. \quad (22)$$

In this case the observed baseband residual contains both the remote FAE contribution and the relative oscillator contribution. FAE is removed only by the denominator/reference in Eq. 22; the baseband residual itself must not be called FFO.

6.6 Dimensional and sign checks

- If $f_{RX}^{(I)} = f_{ref,R}$, Eq. 10 returns zero.
- If the received tone increases while all calibrations are fixed, the reconstructed frequency and FFO must increase.
- Reversing I/Q orientation changes both s_{IQ} and the measured phase-rate sign, leaving $s_{IQ}\widehat{\Delta f}_{BB}$ unchanged.
- Omitting a positive δf_L under the declared convention biases $\widehat{f}_{RX}$ low and therefore biases FFO low.
- Replacing the exact denominator by f_0 gives only a first-order approximation and needlessly introduces a small FAE-dependent error.

7 Derivation from received RF to sampled I/Q

7.1 Passband analytic representation

Over the accepted Mode-0 window, write the desired physical RF signal at the initiator antenna as

$$r_{RF}(t) = \Re \left\{ u(t) e^{j2\pi f_R^{tx,phys}[k]t} \right\} + n_{RF}(t), \quad (23)$$

where $\Re\{\cdot\}$ denotes real part, $u(t) = A(t)e^{j\phi_{ch}(t)}$ is the complex channel envelope, $A(t) \geq 0$ is received amplitude, $\phi_{ch}(t)$ is channel and initial phase, and $n_{RF}(t)$ is additive RF disturbance. For a static narrowband channel, $u(t)$ is approximately constant during the accepted window.

Expressing Eq. 23 in initiator time using Eq. 5 produces a carrier at $f_{RX}^{(I)}[k]$. Let the receiver's calibrated zero-baseband reference be $f_c^{(I)}[k]$. First define a canonical complex downconversion by multiplying the analytic RF signal by $e^{-j2\pi f_c^{(I)}[k]\tau_I}$ and rejecting the high-frequency image. The canonical output rotates at $f_{RX}^{(I)} - f_c^{(I)}$. The reported I/Q stream is either this canonical output ($s_{IQ} = +1$) or its complex conjugate ($s_{IQ} = -1$). Consequently the desired complex envelope is

$$z(\tau_I) = g(\tau_I) e^{j[2\pi s_{IQ}(f_{RX}^{(I)}[k] - f_c^{(I)}[k])\tau_I + \phi_0]} + d(\tau_I) + v(\tau_I), \quad (24)$$

where $g(\tau_I) \geq 0$ is the received magnitude after gain/filtering, d is DC/LO leakage, and v is complex noise. Comparing Eq. 24 with Eq. 6 identifies

$$\Delta f_{BB}[k] = s_{IQ}(f_{RX}^{(I)}[k] - f_c^{(I)}[k]). \quad (25)$$

This is not an additional assumption; it is the definition of the chosen complex-mixer orientation.

7.2 Sampling

At $\tau_n = nT_s = n/F_s$, include residual phase variation $\psi[n]$:

$$z[n] = \rho[n] e^{j(\phi_0 + \omega n + \psi[n])} + d[n] + v[n], \quad \omega = \frac{2\pi \Delta f_{BB}}{F_s}. \quad (26)$$

The units are explicit:

$$\left[\frac{\text{rad}}{\text{sample}} \right] = 2\pi \left[\frac{\text{cycle}}{\text{s}} \right] \left[\frac{\text{s}}{\text{sample}} \right]. \quad (27)$$

In the ideal constant-envelope, no-noise case,

$$z[n] = A e^{j(\phi_0 + \omega n)}, \quad \text{Arg}\{z[n]z^*[n-1]\} = \omega \pmod{2\pi}. \quad (28)$$

Thus the measured phase increment directly gives baseband frequency, subject to alias resolution.

7.3 Average frequency and phase slope

For any differentiable complex tone $z(\tau) = \rho(\tau)e^{j\theta(\tau)}$, instantaneous baseband frequency is

$$f_{BB}(\tau) = \frac{1}{2\pi} \frac{d\theta(\tau)}{d\tau}. \quad (29)$$

Its average over the accepted interval $[\tau_a, \tau_b]$ is

$$\bar{f}_{BB} = \frac{1}{\tau_b - \tau_a} \int_{\tau_a}^{\tau_b} f_{BB}(\tau) d\tau = \frac{\theta(\tau_b) - \theta(\tau_a)}{2\pi(\tau_b - \tau_a)}. \quad (30)$$

The regression and correlation estimators in Section 8 are noise-robust discrete approximations to Eq. 30. They estimate average phase rate, which is exactly the signal-level quantity needed to reconstruct average $f_{RX}[k]$.

7.4 What channel phase does to Mode-0 estimation

A constant channel phase changes ϕ_0 but not ω , so it does not bias frequency. A time-varying channel phase contributes

$$\Delta f_{ch}(\tau) = \frac{1}{2\pi} \frac{d\phi_{ch}(\tau)}{d\tau}. \quad (31)$$

Motion, rapid fading, or switching transients can therefore resemble frequency error. This is why a coherence or phase-linearity test is part of a practical estimator.

7.5 DC and sample-clock cautions

Subtracting the sample mean can remove desired signal when $|\Delta f_{BB}|T_{obs} \ll 1$ cycle. Near zero residual frequency, use a pre-calibrated DC estimate or jointly fit a constant plus a complex sinusoid.

If the estimator uses a programmed rate $F_{s,cmd}$ rather than the calibrated local rate F_s , then

$$\widehat{\Delta f}_{BB,cmd} = \frac{F_{s,cmd}}{F_s} \Delta f_{BB}. \quad (32)$$

This creates a multiplicative frequency error. If the LO and ADC share a reference, their errors may be correlated and must be calibrated as a system rather than added as independent uncertainties.

8 Frequency-estimator derivations and direct implementation

8.1 Lag- m correlation estimator

Begin with the ideal model $z[n] = Ae^{j(\phi_0 + \omega n)}$, where $A > 0$ is a constant sample magnitude. Its lag- m product is

$$z[n]z^*[n-m] = Ae^{j(\phi_0 + \omega n)} Ae^{-j(\phi_0 + \omega(n-m))} \quad (33)$$

$$= A^2 e^{jm\omega}. \quad (34)$$

The unknown initial phase cancels. With real nonnegative weights,

$$R_m \equiv \sum_{n=m}^{N-1} w_n z[n]z^*[n-m] = A^2 e^{jm\omega} \sum_{n=m}^{N-1} w_n. \quad (35)$$

Therefore

$$\text{Arg } R_m = m\omega \pmod{2\pi}. \quad (36)$$

The unwrapped solution is

$$\widehat{\omega}_m = \frac{\text{Arg } R_m + 2\pi q}{m}, \quad q \in \mathbb{Z}, \quad (37)$$

and, using Eq. 26,

$$\widehat{\Delta f}_{BB,m} = \frac{F_s}{2\pi m} (\text{Arg } R_m + 2\pi q). \quad (38)$$

For $q = 0$, the unambiguous interval is

$$-\frac{F_s}{2m} < \Delta f_{BB} \leq \frac{F_s}{2m}. \quad (39)$$

A practical receiver obtains a coarse estimate with $m = 1$, then chooses q for a longer lag by selecting the solution closest to the coarse estimate.

Under zero-mean white complex noise independent across samples, the signal contribution to $\mathbb{E}[R_m]$ retains the phase $m\omega$. Noise increases variance; correlated interference or DC leakage can bias the angle and must be detected or modeled.

8.2 Unwrapped-phase least-squares estimator

Let unwrap denote addition of integer multiples of 2π so that adjacent phase samples follow a continuous branch, and define

$$\theta_n = \text{unwrap}(\arg z[n]), \quad t_n = nT_s. \quad (40)$$

Fit the affine phase model

$$\theta_n = \alpha + \beta t_n + e_n, \quad (41)$$

where α is intercept, $\beta = 2\pi\Delta f_{BB}$ is slope in rad/s, and e_n is phase residual. Minimize

$$J(\alpha, \beta) = \sum_{n=0}^{N-1} w_n (\theta_n - \alpha - \beta t_n)^2. \quad (42)$$

Setting $\partial J/\partial\alpha = 0$ and $\partial J/\partial\beta = 0$ yields

$$\hat{\beta} = \frac{\sum_n w_n (t_n - \bar{t}_w)(\theta_n - \bar{\theta}_w)}{\sum_n w_n (t_n - \bar{t}_w)^2}, \quad (43)$$

with weighted means

$$\bar{t}_w = \frac{\sum_n w_n t_n}{\sum_n w_n}, \quad \bar{\theta}_w = \frac{\sum_n w_n \theta_n}{\sum_n w_n}. \quad (44)$$

The frequency estimate is

$$\widehat{\Delta f_{BB,LS}} = \frac{\hat{\beta}}{2\pi}. \quad (45)$$

The residuals $\hat{e}_n = \theta_n - \hat{\alpha} - \hat{\beta}t_n$ are valuable diagnostics for phase slips, drift, interference, and invalid window edges.

8.3 Variance of the phase-slope estimate

If e_n are independent, zero mean, and have common variance σ_ϕ^2 , and equal weights are used, then

$$\text{var}(\hat{\beta}) = \frac{\sigma_\phi^2}{\sum_n (t_n - \bar{t})^2}. \quad (46)$$

For $t_n = nT_s$,

$$\sum_{n=0}^{N-1} (t_n - \bar{t})^2 = T_s^2 \frac{N(N^2 - 1)}{12}. \quad (47)$$

Therefore

$$\text{var}(\widehat{\Delta f_{BB,LS}}) = \frac{3\sigma_\phi^2}{\pi^2 T_s^2 N(N^2 - 1)}. \quad (48)$$

This simplified result explains the strong benefit of a longer valid observation. It is not a conformance limit and is optimistic when phase noise is correlated or the channel changes.

8.4 FFT bin spacing is not estimator quantization

An N -point DFT has grid spacing

$$\Delta f_{bin} = \frac{F_s}{N} \approx \frac{1}{T_{obs}}. \quad (49)$$

For 80 μs , this is 12.5 kHz. A sinusoid between bins changes the complex DFT values continuously. Interpolation, maximum-likelihood fitting, phase correlation, or regression can estimate a frequency much more finely than one bin. Precision is limited by SNR, duration, phase noise, drift, calibration, and model mismatch.

8.5 Complete per-tone algorithm

For each Mode-0 step k :

1. Locate T_{FM} from the valid reflector synchronization packet and select N accepted samples.
2. Apply validated clipping, DC, and I/Q checks without erasing a near-zero residual tone.
3. Compute R_1 and a coarse $\widehat{\Delta f}_{BB,1}$.
4. Optionally refine with a longer lag or Eq. 45; resolve all aliases against the coarse estimate.
5. Calculate $\widehat{f}_{RX}^{(I)}[k]$ with Eq. 19.
6. Read the reflector's channel-specific $\text{FAE}_R[k]$ and calculate FFO with Eq. 10, equivalently Eq. 21.
7. Store sample bounds, F_s , mixer sign, local calibration, FAE table version, estimator residuals, uncertainty, and validity flags.

End-to-end sign test. Inject RF tones at $f_{ref,R} - \Delta$, $f_{ref,R}$, and $f_{ref,R} + \Delta$. The final FFO results must be negative, approximately zero, and positive. This single test exercises I/Q orientation, NCO sign, LFAE sign, RF reconstruction, and FAE normalization.

9 Combining Mode-0 steps

The Core uses FFO_{RX}^* for an estimate derived from all Mode-0 tones in a subevent but does not prescribe a particular estimator. Let $y_k = \widehat{\text{FFO}}_{RX}[k]$, let $g_k \in \{0, 1\}$ be a validity gate, and let σ_k be an estimated standard deviation. Let $w_{max} > 0$ cap the influence of any one tone, and let $\sigma_{floor} > 0$ prevent unrealistically large weights. One implementation, valid when at least one $g_k = 1$, is

$$w_k = g_k \min \left(w_{max}, \frac{1}{\sigma_k^2 + \sigma_{floor}^2} \right), \quad \widehat{\text{FFO}}_{RX}^* = \frac{\sum_{k=1}^M w_k y_k}{\sum_{k=1}^M w_k}. \quad (50)$$

Do not average $f_{RX}[k]$ in hertz across different channels. First normalize each result by its own $f_0[k](1 + a_R[k])$. Record accepted-tone count, channel span, residual scatter, maximum deviation, and the age of the estimate at each later compensated step.

10 Phase calibration and relation to Mode-0

10.1 Reciprocal PCT phase cancellation

Volume 1, Part A, Sec. 9.2 defines one-way channel phase $\theta_{CH}(f)$ and relative LO phase $\Delta\theta_{LO}(f)$. The measured phases are

$$\theta_{REFL}(f) = \theta_{CH}(f) + \Delta\theta_{LO}(f), \quad (51)$$

$$\theta_{INIT}(f) = \theta_{CH}(f) - \Delta\theta_{LO}(f). \quad (52)$$

With measured amplitudes A_{REFL} and A_{INIT} ,

$$\text{PCT}_{REFL}(f) = A_{REFL}(f)e^{j\theta_{REFL}(f)}, \quad (53)$$

$$\text{PCT}_{INIT}(f) = A_{INIT}(f)e^{j\theta_{INIT}(f)}. \quad (54)$$

Define $H^2(f)$ as the reciprocal PCT product observable and, under the symmetric-channel assumption, define $A_{CH}^2(f) \equiv A_{REFL}(f)A_{INIT}(f)$. Their product is

$$H^2(f) = \text{PCT}_{REFL}(f)\text{PCT}_{INIT}(f) \quad (55)$$

$$= A_{REFL}(f)A_{INIT}(f)e^{j[\theta_{REFL}(f)+\theta_{INIT}(f)]} \quad (56)$$

$$= A_{CH}^2(f)e^{j2\theta_{CH}(f)}. \quad (57)$$

The relative LO phase cancels because it enters with opposite signs. The remaining phase is twice the one-way channel phase.

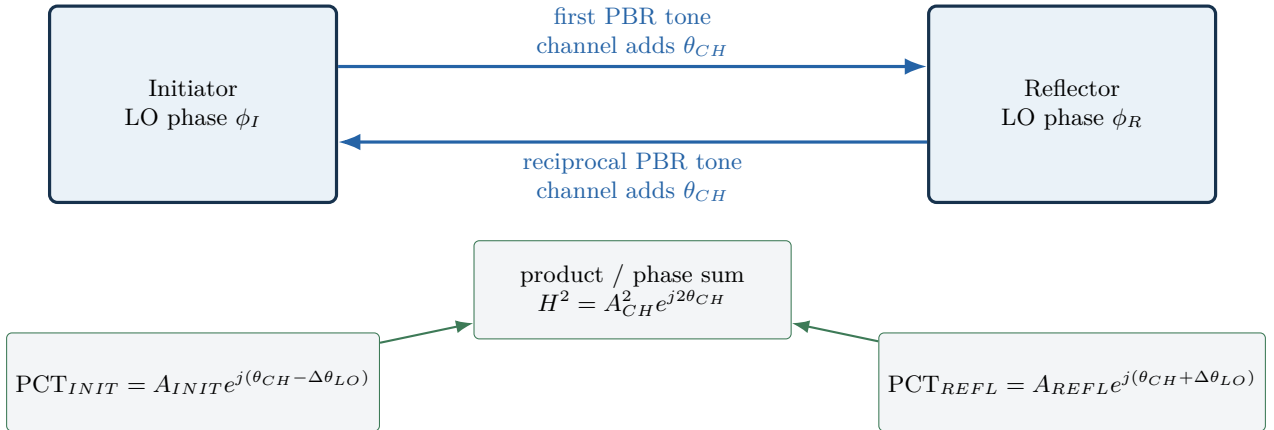


Figure 3: Conventional reciprocal PCT phase calibration. Static relative LO phase cancels in the product.

10.2 Inline PCT Transfer

With IPT enabled, the reflector measures its incoming PCT and adjusts the outgoing tone phase in the PHY. The ideal initiator observable is directly

$$\theta_{INIT,IPT}(f) = 2\theta_{CH}(f). \quad (58)$$

The reflector returns PCT values with zero phase; the real component conveys the defined amplitude and the imaginary component is zero. IPT changes where phase transfer is performed, not the ideal channel-phase observable.

10.3 LFAE phase correction is not Mode-0 FFO substitution

Volume 6, Part A, Sec. 6.3 requires a device aware of local frequency actuation error to compensate the phase rotation that error causes between received and transmitted tone centers. In the nominal form,

$$\angle \text{PCT}' = \angle \text{PCT} - 2\pi\Delta f_k T_{PM_CENTER_DELTA}, \quad (59)$$

where Δf_k is the local actuation error in hertz and $T_{PM_CENTER_DELTA}$ is the relevant antenna-port center separation. Circuit delay and synchronization error may require an implementation-specific correction to the nominal interval.

Part A also states that a device should not compensate a residual frequency error observed during the PCT phase-measurement period except its own LFAE. This is distinct from using Mode-0 FFO for the specified inter-device frequency/timing relationship elsewhere in the CS processing chain.

10.4 Why residual frequency still matters to PBR

If a residual frequency error δf is carried between two phase references separated by Δt , the phase error is

$$\delta\phi_{CFO} = 2\pi\delta f\Delta t. \quad (60)$$

Reciprocal PCT multiplication cancels a static $\Delta\theta_{LO}$ but does not magically remove all time-dependent phase. Correct implementation follows the exact Core correction points: Mode-0 establishes the subevent frequency/timing reference, local LFAE is applied where required, and PCTs represent the specified phase-valid centers.

11 Complete numerical derivation at 2440 MHz

Assume step k has

$$\begin{aligned} f_0[k] &= 2,440,000,000 \text{ Hz}, & \text{FAE}_R[k] &= -1.5 \text{ ppm}, \\ \text{FFO}_{R/I}[k] &= +12 \text{ ppm}, & F_s &= 4 \text{ Msample s}^{-1}, \\ f_{c,cmd}^{(I)}[k] &= f_0[k], & \text{LFAE}_I[k] &= +0.75 \text{ ppm}, \\ s_{IQ} &= +1, & T_{obs} &= 80 \mu\text{s}. \end{aligned}$$

The LFAE sign follows Section 5: actual zero-baseband reference minus commanded reference.

11.1 Remote FAE-adjusted reference and received frequency

Convert ppm to dimensionless values:

$$a_R = -1.5 \times 10^{-6}, \quad \eta_{R/I} = 12 \times 10^{-6}. \quad (61)$$

Then

$$f_{ref,R} = 2,440,000,000(1 - 1.5 \times 10^{-6}) \quad (62)$$

$$= 2,439,996,340 \text{ Hz}, \quad (63)$$

and the exact received frequency in the initiator coordinate is

$$f_{RX}^{(I)} = f_{ref,R}(1 + 12 \times 10^{-6}) \quad (64)$$

$$= 2,440,025,619.956,08 \text{ Hz}. \quad (65)$$

11.2 Local zero-baseband reference including LFAE

The local actuation correction in hertz is

$$\delta f_L^{(I)} = 2,440,000,000(0.75 \times 10^{-6}) = 1,830 \text{ Hz}. \quad (66)$$

Therefore

$$f_c^{(I)} = f_{c,cmd}^{(I)} + \delta f_L^{(I)} = 2,440,001,830 \text{ Hz}. \quad (67)$$

11.3 What the I/Q samples contain

From Eq. 6,

$$\Delta f_{BB} = s_{IQ}(f_{RX}^{(I)} - f_c^{(I)}) \quad (68)$$

$$= +1(2,440,025,619.95608 - 2,440,001,830) \quad (69)$$

$$= 23,789.956,08 \text{ Hz.} \quad (70)$$

At 4 Msamples^{-1} ,

$$\omega = 2\pi \frac{23,789.95608}{4,000,000} = 0.037369 \text{ rad/sample.} \quad (71)$$

For the full tone, $N = F_s T_{obs} = 320$ and the desired phase advances

$$2\pi \Delta f_{BB} T_{obs} = 11.9581 \text{ rad} = 1.90320 \text{ cycles.} \quad (72)$$

11.4 Reconstruction from the measured phase rate

Assume the estimator returns Eq. 70. Equation 19 gives

$$\hat{f}_{RX}^{(I)} = 2,440,000,000 + 1,830 + 23,789.95608 \quad (73)$$

$$= 2,440,025,619.956,08 \text{ Hz,} \quad (74)$$

which matches Eq. 65.

11.5 Final FFO

$$\widehat{\text{FFO}}_{RX} = 10^6 \frac{2,440,025,619.95608 - 2,439,996,340}{2,439,996,340} \quad (75)$$

$$= +12.000000 \text{ ppm.} \quad (76)$$

Two incorrect shortcuts show why the complete derivation matters.

- If remote FAE is ignored and $(f_{RX} - f_0)/f_0$ is reported, the result is about +10.5 ppm, not +12 ppm.
- If the positive local LFAE correction is omitted, the reconstructed frequency is low by 1,830 Hz and the FFO result is biased low by approximately 0.750001 ppm.

11.6 Residual frequency translated to phase

The +12 ppm component relative to $f_{ref,R}$ is 29,279.956,08 Hz. If left uncompensated across $20 \mu\text{s}$, it produces

$$2\pi(29,279.95608)(20 \times 10^{-6}) = 3.679 \text{ rad} = 210.8^\circ. \quad (77)$$

This numerical result explains why frequency calibration and phase calibration are separate but coupled requirements.

12 Error sources, diagnostics, and verification

Table 4: Representative error budget.

Source	Mathematical effect	Mitigation	Diagnostic
Thermal noise	Random error in $\text{Arg } R_m$ or θ_n	Use valid duration and efficient weighting	Correlation magnitude and repeated variance
Phase noise	Correlated $\psi[n]$	Validate estimator bandwidth	Residual spectrum
Within-tone drift	Curvature in phase	Fit curvature or reject transient region	Quadratic phase term
RF/AGC settling	Biased early samples	Characterized t_{settle}	Start-index sweep
Ramp boundary	Biased late samples	Characterized t_{tail}	End-index sweep
DC/LO leakage	Pull toward zero phase rate	Calibrated DC or joint model	DC-to-tone ratio
I/Q imbalance	Image tone and phase ripple	Gain/quadrature calibration	Image rejection
Sample-rate error	Scales $\Delta \hat{f}_{BB}$	Use calibrated F_s	Digital tone injection
LFAE sign/value	Additive error in $f_c^{(I)}$	Declare actual-minus-command sign	Three-point RF test
Remote FAE mismatch	Wrong denominator/reference	Use correct channel table entry	Compare table version
Alias selection	Adds qF_s/m	Coarse-to-fine estimator	Lag consistency
Motion/multipath	Adds $d\phi_{ch}/dt$	Coherence and residual gates	Window and antenna consistency
PCT center error	Phase term $2\pi\delta f\Delta t$	Use specified valid center	Center-time sweep

12.1 Required verification sequence

1. Generate noise-free complex tones and verify Eqs. 34-38 over positive and negative frequency.
2. Add known phase noise, DC, I/Q imbalance, clipping, and drift; verify bias and quality gates.
3. Inject a digital tone after downconversion to verify F_s , decimation, NCO bookkeeping, and fixed-point scaling.
4. Conducted RF sweep at $f_{ref,R} - \Delta$, $f_{ref,R}$, and $f_{ref,R} + \Delta$ to verify the final sign.
5. Repeat over CS channel, input level, temperature, supply, synchronization PHY, and supported T_{IP1} .
6. Use several Mode-0 steps and controlled outliers to verify channel-specific FAE selection and aggregation.
7. Inject known reciprocal PCT phases to verify LO-phase cancellation and the factor of two.
8. Compare IPT enabled and disabled after the prescribed processing; both should yield the same ideal $2\theta_{CH}$ within uncertainty.

13 Normative behavior versus implementation choice

Bluetooth-defined behavior	Implementation-specific design
Mode-0 exchange order and timing intervals	RF/IF architecture, sample rate, and numeric format
Meaning of f_0 , f_{RX} , FAE, LFAE, FFO, and FFO_{RX}^*	Definition of the internal zero-baseband reference and calibration storage
Average received tone frequency as the measurement target	Correlation, regression, FFT, maximum-likelihood, or hybrid estimator
Use of local actuation error compensation	Sign convention and location of the correction in the receiver chain
PCT valid region and LFAE phase correction	Sample weights and quality metrics inside the valid region
Reciprocal PCT product and IPT behavior	Hardware phase rotation and internal PCT estimator

14 Conclusions

A self-contained Mode-0 implementation requires one declared local frequency coordinate. The reflector transmits a tone whose physical frequency contains its FAE-adjusted reference and its clock-scale error. The initiator observes phase rotation relative to its actual zero-baseband frequency. After converting radians per sample with the calibrated sample rate, it reconstructs $f_{RX}[k]$ by adding the commanded zero frequency, the signed local actuation correction, and the signed baseband residual. Only then is the exact FAE-adjusted Bluetooth FFO equation applied.

The direct result is Eq. 21, which links measured I/Q correlation phase to $\widehat{\text{FFO}}_{RX}[k]$ without an unexplained intermediate quantity. The oscillator-frame derivation, Eq. 14, shows why the result is a remote-to-local frequency ratio rather than an absolute laboratory offset. The 2,440 MHz example verifies every sign and intermediate value.

Mode-0 frequency/time calibration does not replace Volume 1 phase calibration. Conventional reciprocal PCT multiplication or IPT cancels the static relative LO phase and exposes $2\theta_{CH}$. Correct PBR processing must also use the specified phase-valid centers and the required LFAE phase correction.

A Compact implementation equations

For a validated lag- m estimator:

$$R_m[k] = \sum_{n=m}^{N-1} w_n z_k[n] z_k^*[n-m], \quad (78)$$

$$\widehat{\Delta f}_{BB}[k] = \frac{F_s}{2\pi m} (\text{Arg } R_m[k] + 2\pi q_k), \quad (79)$$

$$\widehat{f}_{RX}^{(I)}[k] = f_{c,cmd}^{(I)}[k] + \widehat{\delta f}_L^{(I)}[k] + s_{IQ} \widehat{\Delta f}_{BB}[k], \quad (80)$$

$$\widehat{\text{FFO}}_{RX}[k] = 10^6 \left[\frac{\widehat{f}_{RX}^{(I)}[k]}{f_0[k](1 + 10^{-6} \text{FAE}_R[k])} - 1 \right] \text{ ppm}. \quad (81)$$

These four equations are sufficient only when their symbols follow Section 5, aliases are resolved, F_s is calibrated, and the sample window lies inside T_{FM} .

B Numerical test vector

Table 5: Deterministic values for the 2,440 MHz example.

Quantity	Value	Unit
f_0	2,440,000,000	Hz
Reflector FAE	-1.5	ppm
$f_{ref,R}$	2,439,996,340	Hz
Relative FFO	+12	ppm
$f_{RX}^{(I)}$	2,440,025,619.95608	Hz
Commanded zero frequency	2,440,000,000	Hz
Initiator LFAE	+0.75	ppm
$\delta f_L^{(I)}$	+1,830	Hz
Actual zero-baseband reference	2,440,001,830	Hz
Δf_{BB} for $s_{IQ} = +1$	23,789.95608	Hz
F_s	4,000,000	sample/s
N over 80 μ s	320	samples
Phase increment ω	0.037369	rad/sample
Corrected FFO	+12.000000	ppm

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